

PROGRESO, Mexico

WMO: 765930

Lat: 21.2767N

Lon: 89.6539W

Elev: 2

StdP: 101.30

Time Zone: -6.00 (MEX)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	18.6	12.1	8.8	20.5	13.4	9.6	21.1	11.4	22.6	9.2	22.6	2.0	140	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	34.6	27.8	33.3	27.3	32.2	27.3	29.9	32.6	29.2	31.7	28.7	31.1	4.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.2	25.9	32.3	28.5	24.9	31.4	28.0	24.1	30.8	99.1	32.7	95.5	31.8	92.8	31.1	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.3	7.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	23.2	24.2	25.4	26.9	27.7	28.0	27.8	27.9	27.7	27.1	25.4	24.3
	DBStd	2.30	1.93	1.90	2.32	2.14	1.52	1.27	0.89	1.01	0.99	1.25	1.69	1.66
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5956	408	398	479	506	550	541	552	554	532	531	463	443
	CDD18.3	2914	149	165	221	256	292	291	293	296	282	273	213	184
	CDH23.3	27954	635	991	1792	2528	3221	3266	3352	3348	3176	2871	1728	1047
CDH26.7	8504	66	197	521	814	1113	1179	1154	1167	1031	807	302	154	
Wind	WSAvg	3.4	3.0	3.5	4.3	3.6	3.8	3.2	3.5	3.3	2.8	3.1	3.8	3.4
Precipitation	PrecAvg	412	19	17	8	9	19	81	46	54	73	57	27	21
	PrecMax	742	118	94	70	79	83	293	118	181	215	232	123	125
	PrecMin	103	0	0	0	0	0	0	0	10	0	0	0	0
	PrecStd	173	27	26	16	17	21	76	32	47	61	60	32	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	33.1	34.9	36.6	35.8	35.3	34.2	34.5	33.9	33.4	31.6	31.7
		MCWB	23.7	24.2	27.8	28.7	27.6	28.5	28.6	28.3	29.8	28.2	27.4	24.5
	2%	DB	28.5	31.0	32.8	34.0	33.6	33.3	32.5	32.5	31.7	31.7	30.2	29.8
		MCWB	23.3	24.4	26.3	26.5	27.3	27.3	28.4	28.1	28.3	27.4	26.4	25.3
	5%	DB	27.0	29.3	31.1	32.1	31.9	31.8	31.6	31.5	31.0	30.8	29.2	28.1
		MCWB	23.1	24.2	25.7	25.9	27.1	27.4	28.1	27.9	28.0	27.1	25.7	24.4
	10%	DB	26.1	27.9	29.3	30.5	30.8	30.8	30.8	30.9	30.2	30.0	28.4	27.0
		MCWB	22.9	23.9	25.2	25.5	26.7	27.3	27.7	27.7	27.3	26.6	25.3	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	27.5	29.6	31.0	30.6	30.3	30.5	29.9	30.1	29.4	29.0	26.8
		MCDB	28.1	30.9	33.7	35.0	34.4	33.3	33.1	32.4	32.7	31.9	31.0	29.4
	2%	WB	24.2	26.0	28.2	28.2	28.9	29.1	29.5	29.1	29.1	28.5	27.5	25.7
		MCDB	26.8	28.5	31.6	31.6	31.6	31.5	31.7	31.5	31.4	30.8	29.3	28.1
	5%	WB	23.7	25.1	26.9	26.9	27.9	28.3	28.7	28.5	28.4	27.6	26.5	25.1
		MCDB	26.3	27.5	29.4	30.5	30.8	30.6	30.9	31.0	30.6	30.0	28.3	27.2
	10%	WB	23.3	24.2	25.9	26.1	27.1	27.5	28.0	27.8	27.7	26.9	25.7	24.4
		MCDB	25.7	26.8	28.3	29.4	30.2	30.0	30.4	30.4	29.8	29.3	27.7	26.6
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	7.7	7.2	6.5	6.1	5.7	5.7	5.0	5.3	5.5	5.7
		MCDBR	7.5	10.0	11.0	10.7	9.2	8.3	8.2	7.9	7.0	7.2	6.9	8.0
		MCWBR	4.0	4.9	5.6	5.2	4.7	4.3	4.9	4.2	4.4	3.9	3.9	4.1
	5% WB	MCDBR	5.9	8.7	10.0	10.0	8.4	7.6	7.8	7.3	7.4	7.0	5.9	6.6
		MCWBR	3.8	5.3	6.6	5.9	5.2	4.9	5.4	4.8	5.0	4.5	4.1	4.2
Clear-Sky Solar Irradiance	taub		0.362	0.376	0.401	0.436	0.451	0.449	0.465	0.442	0.411	0.397	0.376	0.373
	taud		2.565	2.520	2.417	2.299	2.310	2.373	2.308	2.391	2.462	2.506	2.525	2.537
	Ebn at Noon		913	918	904	872	850	845	831	855	882	885	890	887
	Edh at Noon		93	103	118	135	132	123	132	121	112	103	96	93
All-Sky Solar Radiation	RadAvg		4.10	4.95	5.86	6.48	6.61	6.10	6.61	6.45	5.80	5.10	4.43	3.99
	RadStd		0.30	0.39	0.27	0.31	0.32	0.35	0.24	0.20	0.30	0.23	0.21	0.25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.88	+1.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page